

Matrix inequalities and norms

Bourin's crossed-Heinz inequality

Difficulty: extreme **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: A longstanding sharp bound for all unitarily invariant norms remains beyond the known interpolation range; it would strengthen matrix-power and matrix-mean estimates.

Problem statement

For every integer $n \geq 1$, positive definite $A, B \in \mathbb{C}^{n \times n}$, real $t \in [0, 1]$, and unitarily invariant norm $\|\cdot\|$, is

$$\|A^t B^{1-t} + B^t A^{1-t}\| \leq \|A + B\|?$$

Powers use the spectral functional calculus. A norm is unitarily invariant when $\|UXV\| = \|X\|$ for all unitary U, V . The positive definite formulation avoids ambiguity at exponent zero; for $0 < t < 1$ it includes the positive semidefinite case by continuity.

Why it matters

This asks for a sharp bound on noncommuting products of fractional matrix powers. Bounds of this kind support analysis of positive definite matrix computations and the comparison of matrix means.

References

1. J.-C. Bourin, *Matrix subadditivity inequalities and block-matrices*, International Journal of Mathematics 20(6) (2009), 679–691; the crossed-Heinz question. [Preprint](#), [DOI](#).
2. F. Kittaneh and É. Ricard, *On a question of Bourin*, Linear Algebra and its Applications 710 (2025), 356–362; central parameter interval. [DOI](#).
3. T. Zhang, *Bourin-type inequalities for τ -measurable operators in fully symmetric spaces*. arXiv:2602.00358v1 (30 January 2026), §1, equation (1.3), Theorem 1.3. [Primary text](#).

Status check — 2026-09-10

The bound with constant one is proved for $t \in [1/4, 3/4]$; the endpoints 0, 1 are immediate for positive definite matrices. The displayed assertion remains unresolved in the two outer open intervals. The latest 2026 source gives a larger explicit constant there. Searches included *Bourin crossed Heinz conjecture 2026*, *On a question of Bourin Kittaneh Ricard*, and *Bourin inequality proof counterexample 2025 2026*. No resolution of the remaining norm inequality was found. The stronger assertion for individual singular values has counterexamples and is not the present problem.

Audit update (2026-09-10): Rechecked Zhang's Theorem 1.3 and searched for a later crossed-Heinz proof or counterexample. Constant one is established on $[1/4, 3/4]$; the larger constant on the outer intervals leaves the stated question there unresolved. This is a bounded literature check, not a proof that no solution exists.